

Exercise: Permutation Tests and Bootstrap Confidence Intervals for Self-made Statistics

Advanced Statistics for Neuroscience | Matlab exercise

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Overview

In neurobiology, experimental effects often do not appear as a simple shift of the mean. A manipulation may affect only a subpopulation of neurons, synapses, spines, cells, or pixels, for example by increasing the number of strongly responding cells or unusually large synaptic structures.

In such cases, it can be more informative to define a statistic that directly captures the biological question of interest, and then use permutation testing and bootstrapping to evaluate this statistic.

In this exercise, you will generate two synthetic samples, WT and KO, with $N = 1000$ observations each. You will define two self-made statistics, test them with permutation tests, and estimate their confidence intervals with bootstrap resampling.

Generate observations

Write a Matlab or Python programme that generates two samples, WT and KO, with $N = 1000$ observations in each group.

$WT \sim N(0, 1)$

The KO sample should mostly be drawn from the same distribution, but should contain a small fraction of high-response observations:

$KO \text{ baseline responses} \sim N(0, 1)$

$KO \text{ high responses} \sim N(3, 0.5)$

For example, let 10% of KO observations come from the high-response distribution.

Discussion

- Why does this data-generating process create a difference in the upper tail of the distribution?
- Would this difference necessarily be well described by a change in the mean?

Visualize your samples

Plot the WT and KO observations. Use at least two of the following visualizations:

- histogram
- cumulative distribution plot
- scatter / strip plot
- boxplot

Discussion

- Do WT and KO differ across the whole distribution, or mainly in the upper tail?
- Is the KO distribution symmetric?
- Can you visually detect the high-response subpopulation?

First statistic: difference in skewness

Compute the skewness of WT and KO and define

$T_skew = skewness(KO) - skewness(WT)$

This statistic asks whether KO has a stronger right tail than WT.

Discussion

- What biological question does this statistic represent?
- Why might skewness be more informative than the mean in this example?
- What kind of distributional difference would make T_skew large and positive?

Second statistic: upper-tail enrichment

Pool all WT and KO observations and compute the 80th percentile of the pooled distribution:

```
q80 = 80th percentile of [WT, KO]
```

Then define

```
T_tail = fraction(KO > q80) - fraction(WT > q80)
```

This statistic asks whether KO contributes more observations to the upper tail of the pooled distribution.

Discussion

- What biological question does this statistic represent?
- Why might this statistic be useful for detecting a small subpopulation of strongly responding cells?
- Why is this statistic not equivalent to comparing group means?

Permutation test

Pool all observations from WT and KO. Randomly shuffle the group labels many times. In every permutation, create two new groups of size $N = 1000$ and compute both statistics:

```
T_skew_perm
```

```
T_tail_perm
```

Use, for example, $N_{\text{perm}} = 10000$ permutations. This gives two null distributions:

- Null distribution for T_{skew}
- Null distribution for T_{tail}

Plot both null distributions as histograms and mark the observed statistic in each plot.

Discussion

- Is the observed statistic near the centre of the null distribution or in one of its tails?
- Do both statistics lead to the same conclusion?
- Which statistic seems better matched to the way the data were generated?

Permutation p-values

Compute one-sided permutation p-values for the hypothesis that KO has a stronger right tail than WT:

```
p_skew = fraction(T_skew_perm >= T_skew_observed)
```

```
p_tail = fraction(T_tail_perm >= T_tail_observed)
```

Optionally, also compute two-sided p-values:

```
p = fraction(abs(T_perm) >= abs(T_observed))
```

Discussion

- Which statistic gives the smaller p-value?
- Why can two different self-made statistics give different p-values for the same data?
- Which p-value would you report, and why?

Bootstrap confidence intervals

Now estimate the uncertainty of both observed statistics using bootstrap resampling. Randomly resample WT with replacement and KO with replacement. Each bootstrap sample should again contain $N = 1000$ observations per group.

For every bootstrap repetition, compute

```
T_skew_boot
```

```
T_tail_boot
```

Use, for example, $N_{\text{boot}} = 2000$ bootstrap repetitions.

Estimate 95% confidence intervals as the 2.5th and 97.5th percentiles of the bootstrap distributions:

```
CI_skew = [2.5th percentile, 97.5th percentile] of T_skew_boot
```

$CI_{tail} = [2.5th \text{ percentile}, 97.5th \text{ percentile}] \text{ of } T_{tail_boot}$

Plot the bootstrap distributions for both statistics and mark the observed values and confidence intervals.

Discussion

- Is zero inside the bootstrap confidence interval?
- Are the confidence intervals narrow or wide?
- Do the bootstrap confidence intervals agree with the permutation p-values?
- What is the conceptual difference between a permutation null distribution and a bootstrap sampling distribution?

Repeat with different effect sizes

Repeat the analysis for different fractions of high-response KO observations:

p_high	T_skew	p_skew	95% CI for T_skew	T_tail	p_tail	95% CI for T_tail
0.02						
0.05						
0.10						
0.20						

Report the observed statistic, permutation p-value, and bootstrap confidence interval for each value of p_high.

Discussion

- Which statistic becomes significant first as p_high increases?
- Which statistic gives the more stable confidence interval?
- What happens when the high-response subpopulation is very rare?

Final checklist

1. A short description of the simulated WT and KO data.
2. Plots visualizing the two samples.
3. The observed value of T_skew.
4. The observed value of T_tail.
5. Permutation null distribution for T_skew.
6. Permutation null distribution for T_tail.
7. One-sided permutation p-value for T_skew.
8. One-sided permutation p-value for T_tail.
9. Bootstrap distribution and 95% CI for T_skew.
10. Bootstrap distribution and 95% CI for T_tail.
11. A table showing results for different values of p_high.
12. A short explanation of which statistic is more useful for detecting rare high-response observations.
13. A short explanation of the difference between permutation testing and bootstrapping.

Main conceptual takeaway

Permutation test:

Would a statistic this large appear if WT/KO labels were arbitrary?

Bootstrap:

How precisely have we estimated this statistic from the available sample?

The key idea is that permutation testing does not require a standard named test. It requires a clearly defined statistic, an exchangeable null hypothesis, and a rule for generating the null distribution by shuffling labels.